

Methodology Note -- Team 5guys1repo -- PowerNext-AI 2026 Screening Round

1. Approach

Preprocessing (M1). We load the participant workbook (``Training_Data` / `Test_Data``) with a CSV fallback, freeze exact headers in ``schema.py``, impute with train-fitted medians (sensors only -- the sole missing columns), and deduplicate training rows only (12 dropped, keep-first). All 350 test rows are preserved in input order. We add 37 ``feat_*` columns: power/energy/Joule-heating proxies, S1-S3 consensus (mean, spread, pairwise gaps, per-sensor deviations), S4-relevance probes, rise-per-input ratios, and pre-imputation ``feat_missing_*` flags -- 45 model features total, with ID/targets excluded by construction.

Validity detection (M2). A supervised HistGradientBoosting classifier on engineer labels (StratifiedKFold x5, seed 42, balanced classes) beats a rule baseline and RandomForest on out-of-fold Invalid-F1 (0.925; P 1.00 / R 0.86). The decision threshold (0.15) is tuned on OOF scores. ``Anomaly_Score`` is exactly $P(\text{Invalid})$ in $[0,1]$; ``Reason_Flags`` (trio disagreement, sensor deviation, missingness, duplicates) are deterministic diagnostics. 38/350 test rows are flagged Invalid.

Reference prediction (M3). HistGradientBoosting regression trained on Valid rows only (866), plain KFold x5: OOF MAE 0.67, RMSE 1.07, R^2 0.990 -- far above the linear floor (RMSE 3.98). A Valid-only-vs-all-rows experiment (both scored on Valid held-out folds) keeps Valid-only; an S4 ablation drops the S4 family. Predictions are clipped at 0 at source and asserted finite; test range 13.1-59.4 sits inside the training range (11.9-61.6).

Integration (M4). ``python run_pipeline.py`` reproduces everything from the raw xlsx. The writer renames ``Predicted_Validity`?` ``Validity_Label``, drops diagnostics, and asserts 350/350 rows in input order with finite, non-negative predictions. ``summary.json`` is fully code-generated (template + fitted numbers).

2. Important parameters

Load_Current_A dominates (0.92 permutation importance -- current drives heating), refined by sensor consensus (S2 most trusted, corr 0.70 with target; S1 0.54, S3 0.48), absolute-temperature views, and missingness flags. Sensor_S4 is the planted decoy (corr 0.002; ablation confirms dropping it helps). Voltage, duration, and ambient enter through power/energy interaction terms.

3. Abnormal-data method

We deliberately do NOT equate outlier with invalid: the classifier learns from real labels which unusual patterns engineers accepted. Evidence: 0% OOF false-positive rate on high-current-but-Valid edge rows, e.g. TRN-0762 (110 A, $\geq p99$) scores 0.0006 -- correctly kept Valid. Flags explain each verdict without overriding the supervised signal.

4. Assumptions

Training on Valid rows only is our engineering decision (Invalid rows carry unreliable measurements), disclosed not mandated. Median imputation fitted once on train leaks negligibly. Summary min/max/avg cover ALL predicted rows. Threshold 0.15 and feature medians transfer to the second unseen dataset, which we assume shares the schema; edge-slice OOF checks (high-current RMSE 1.21 vs 1.07 overall) are our robustness proxy.

5. Digital Twin Automation Steps

1. **Connect:** stream test-bench logs into a validated schema service (this pipeline's loader + `schema.py`, rising to a message queue in production).
2. **Preprocess live:** apply frozen medians/features per record; quarantine schema violations for engineer review instead of crashing.
3. **Twin inference:** serve the validity + RP models behind a versioned API; log every score, flag, and prediction with model version.
4. **Attention queue:** rank live records with the §1 attention blend so engineers review the riskiest tests first.
5. **Feedback loop:** record engineer Valid/Invalid verdicts as new labels.
6. **Retrain & gate:** nightly retraining with the frozen CV scheme; promote only on better OOF metrics plus the regime-edge checks.
7. **Monitor:** track input drift, invalid-rate shifts, and prediction-range excursions with alerts; keep deterministic reruns for audit.